

Herbal constituent sequoyitol improves hyperglycemia and glucose intolerance by targeting hepatocytes, adipocytes, and β -cells

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insulin resistance; diabetes; hyperglycemia

GLUCOSE IS A PRIMARY METABOLIC fuel. Blood glucose levels are maintained within a narrow range through the action of insulin and other metabolic hormones (18, 34). Insulin decreases blood glucose by stimulating glucose uptake into adipose tissue and skeletal muscle and by suppressing hepatic glucose production (34). Insulin resistance is the primary risk factor for type 2 diabetes (42). Additionally, pancreatic β -cell function is also impaired in type 2 diabetes and unable to secrete sufficient insulin to compensate for insulin resistance (26). Relative insulin deficiency, due to a combination of insulin resistance and impaired pancreatic β -cell function, contributes to hyperglycemia and glucose intolerance in type 2 diabetes. Small

molecules with the capability to improve insulin sensitivity and/or β -cell function have a great therapeutic potential for the treatment of type 2 diabetes.

Growing evidence indicate that herbal constituents hold a great promise for the treatment of type 2 diabetes, and extracts of numerous plants have been reported to reduce blood glucose (6, 8, 17, 22, 41). In search for small compounds with an anti-diabetic property, we examined natural compounds in plant extracts, focusing on inositol derivatives. Inositols consist of nine isomeric forms (*myo*-, *scyllo*-, *epi*-, *allo*-, *cis*-, *neo*-, *muco*-, *D-chiro*-, *L-chiro*-inositol) and are synthesized from glucose-6-phosphate, a glycolytic metabolite (28). Many forms of inositols and inositol derivatives are essential components of membrane phospholipids in animals and humans (28, 39). These phospholipids not only play an important structural role, but also mediate cell signaling (11, 39, 40). Interestingly, insulin stimulates production of inositol phosphoglycans (IPGs) (21, 29, 36). IPG is believed to act as an intracellular second messenger to mediate insulin metabolic action (21, 23). Moreover, *chiro*-inositol levels are decreased in type 2 diabetes (3), and increased *myo*-inositol-to-*chiro*-inositol ratios are associated with type 2 diabetes and insulin resistance in both animals and humans (20, 23). *D-Chiro*-inositol and 3-*O*-methyl-*D-chiro*-inositol (*D*-pinitol) have been reported to act as an insulin mimetic to improve hyperglycemia in both mice and humans with type 2 diabetes (4, 7, 10, 19, 35). It is important to identify additional inositol derivatives, to evaluate their therapeutic potentials, and to elucidate the cellular mechanisms of their actions.

In this study, we have identified 5-*O*-methyl-*myo*-inositol (sequoyitol) as a new natural compound with antidiabetic properties. Sequoyitol is a herbal constituent. Both oral and subcutaneous administrations of sequoyitol ameliorate hyperglycemia and glucose intolerance in *ob/ob* mice with insulin resistance. Chronic sequoyitol treatments also improve hyperglycemia and glucose intolerance in streptozotocin (STZ)-treated mice with insulin deficiency. Additionally, sequoyitol directly improves insulin sensitivity in cultured hepatocytes, adipocytes, and β -cells, and protects β -cells against oxidative injury. These results suggest that sequoyitol has a therapeutic potential for the treatment of diabetes.

RESEARCH DESIGN AND METHODS

Animals. *ob/ob* and C57BL/6 mice were purchased from the Jackson laboratory (Bar Harbor, ME). Mice were housed on a 12:12-h light-dark cycle in the Unit for Laboratory Animal Medicine at the University of Michigan or in the animal facility at the Institute for Nutritional Sciences, Shanghai Institutes for Biological Sciences,

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Chinese Academy of Sciences, Shanghai, China. Mice were fed a normal chow ad libitum with a free access to water. Animal experiments were conducted following the protocols approved by the University Committee on the Use and Care of Animals at the University of Michigan Medical School and by the Institutional Animal Care and Use Committees at the Institute for Nutritional Sciences, Shanghai Institutes for Biological Sciences, Chinese Academy of Sciences.

Sequoyitol administration, glucose tolerance tests, and insulin tolerance tests. For oral administration, *ob/ob* mice (8–9 wk) received sequoyitol (98% purity, Xiangbei Welman Pharmaceutical, Changsha, China) by gastric gavage (40 mg/kg body wt, twice daily). For subcutaneous administration, *ob/ob* mice (10 wk) were anesthetized with 2–4% isoflurane, and osmotic minipumps (model 2002; Alzet, Cupertino, CA) were implanted subcutaneously. Minipumps were prefilled with either a sterile 0.9% NaCl vehicle or sequoyitol. A sequoyitol-filled pump released sequoyitol at 0.5 nmol/h. For STZ treatments, C57BL/6 males (9 wk) were intraperitoneally injected with STZ (80 mg/kg body wt once a day for 2 days). The mice were fed either tap water (control) or water supplemented with sequoyitol (7 mg/ml). Blood samples were collected from tail veins, and blood glucose and plasma insulin were measured as described previously (31). For glucose tolerance test (GTT), mice were fasted overnight (or 6 h) and intraperitoneally injected with D-glucose (0.6–0.8 g/kg body wt), and blood glucose levels were monitored after glucose injection. For insulin tolerance test (ITT), mice were fasted for 5–6 h and intraperitoneally injected with human insulin, and blood glucose levels were monitored after glucose injection.

Immunoprecipitation and immunoblotting. Mice were fasted overnight, anesthetized, and administered with insulin via inferior vena. Livers were homogenized in a lysis buffer (50 mM Tris-HCl, pH 7.5, 1.0% Nonidet P-40, 150 mM NaCl, 2 mM EGTA, 1 mM Na₃VO₄, 100 mM NaF, 10 mM Na₄P₂O₇, 1 mM PMSF, 10 µg/ml aprotinin, 10 µg/ml leupeptin), as described previously (32, 44). Liver extracts were immunoprecipitated and immunoblotted with the indicated antibodies. Liver extracts were also immunoblotted with antibodies against phospho-Akt pSer⁴⁷³ (Cell Signaling) and pThr³⁰⁸ (Santa Cruz) or total Akt (Santa Cruz). HepG2 cells, 3T3-L1 adipocytes, and INS-1 cells were treated with the indicated ligands. Cell extracts were prepared and subjected to immunoprecipitation and immunoblotting assays.

Adipocyte glucose uptake. 3T3-L1 preadipocytes were differentiated into adipocytes, as described previously (9). The adipocytes were incubated with sequoyitol for 6 or 12 h and subjected to glucose uptake assays using 2-[³H]deoxy-D-glucose (9). C57BL/6 males were fed a high-fat diet (45% fat) for 8 wk, and epididymal fat depots were isolated, minced, and digested in 1 mg/ml type II collagenase (Worthington Biochemicals, Lakewood, NJ) at 37°C for 40 min. The adipocytes were dispersed, filtered through two layers of cloth, washed three times with Krebs-Ringer buffer (130 mM NaCl, 5 mM KCl, 1.3 mM CaCl₂, 1.3 mM MgSO₄, 25 mM HEPES, pH 7.4) containing 4% bovine serum albumin (BSA), and cultured in Krebs-Ringer buffer with 4% BSA. Primary adipocytes were treated with sequoyitol (100 µM) for 3 h and stimulated with 5 nM insulin for 30 min; 2-[³H]deoxy-D-glucose was added during the last 10-min incubation. To stop glucose uptake, adipocytes were mixed with dinonyl phthalate oil (Sigma-Aldrich, St Louis, MO) and centrifuged at 10,000 rpm for 1 min. Adipocytes were on the top of the organic phase and transferred into a scintillation vial and solubilized in 0.5% SDS buffer. 2-[³H]deoxy-D-glucose uptake was determined by scintillation counting and normalized to total adipocyte numbers.

Primary hepatocyte glucose production. Primary hepatocytes were prepared by liver perfusion with type II collagenase, grown on collagen-coated plates, and subjected to glucose production assays, as described previously (9, 44). Briefly, primary hepatocytes were pretreated with sequoyitol (100 µM) or vehicles overnight and subsequently incubated for 4 h in an assay buffer (118 mM NaCl, 2.5 mM

CaCl₂, 4.8 mM KCl, 25 mM NaHCO₃, 1.1 mM KH₂PO₄, 1.2 mM MgSO₄, 10 µg ZnSO₄, 0.6% BSA, 10 mM HEPES, 5 mM lactate, 5 mM pyruvate, and pH 7.4). To increase glucose production, cells were treated with 10 µM N⁶,2'-O-dibutyryl adenosine 3',5'-cyclic monophosphate sodium salt (DB-cAMP) and 100 nM dexamethasone; to decrease glucose production, insulin was added into the assay buffer. Glucose in culture medium was measured and normalized to total protein levels, and the normalized values were used as an index to estimate glucose production.

INS-1 cell viability. Rat INS-1 cells were grown in RPMI-1640 supplemented with 10% FBS, 10 mM HEPES, 2 mM L-glutamine, 1 mM sodium-pyruvate, and 0.05 mM β-mercaptoethanol, as described previously (14). Cells were pretreated with or without sequoyitol (0, 5, or 10 mg/ml) for 6 h and then treated with STZ (0.5 mM) or H₂O₂ (10 µM) in the presence or absence of sequoyitol for an additional 12 h. Cell viability was measured using 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide (MTT) assays.

Statistical analysis. Data are presented as means ± SE. Differences between groups were analyzed by two-tailed Student's *t*-test. *P* < 0.05 was considered statistically significant.

RESULTS

Subcutaneous administration of sequoyitol improves insulin resistance and glucose intolerance in *ob/ob* mice. We examined the antidiabetic effect of herbal extracts and identified sequoyitol as an insulin sensitizer. Sequoyitol is a 5-*O*-methyl derivative of *myo*-inositol (Fig. 1A). To determine whether sequoyitol reduces blood glucose under insulin-resistant conditions, it was subcutaneously administered into *ob/ob* male mice (10 wk) via osmotic minipumps (0.5 nmol/h) for 23 days. The *ob/ob* mice are deficient of leptin and commonly used as a genetic model of obesity and type 2 diabetes. Sequoyitol treatments did not alter body weights (Fig. 1B). Fourteen days after treatments, blood glucose levels were significantly lower in sequoyitol-treated than in vehicle-treated mice (Fig. 1C). Sequoyitol treatments also significantly reduced plasma insulin levels (by 29%) (Fig. 1C). Homeostasis model of assessment index, a commonly used parameter to estimate insulin sensitivity, was reduced by 44% in sequoyitol-treated mice than in vehicle-treated mice (saline: 76.1 ± 6.5, *n* = 6; sequoyitol: 42.3 ± 16.0, *n* = 7; *P* = 0.0006). To further analyze insulin sensitivity and glucose metabolism, we performed GTTs (17 days after treatments) and ITTs (10 days after treatments). Sequoyitol treatments significantly reduced blood glucose levels 30 min after glucose injection during GTT, and area under the curve decreased by 34% in sequoyitol-treated mice (Fig. 1D). Sequoyitol treatments also increased the ability of insulin to reduce blood glucose levels, and area under the curve decreased by 15% in sequoyitol-treated mice (Fig. 1E). These results demonstrate that chronic sequoyitol treatments ameliorate hyperglycemia, hyperinsulinemia, insulin resistance, and glucose intolerance in mice with obesity and insulin resistance.

Subcutaneous administration of sequoyitol enhances insulin signaling in *ob/ob* mice. To determine whether sequoyitol treatments enhance insulin signaling, we examined insulin-stimulated phosphorylation of insulin receptors (IR), IR substrate-1 (IRS1), Akt (pSer⁴⁷³ and pThr³⁰⁸), and ERK1/2 in the liver. The *ob/ob* mice were chronically treated with saline or sequoyitol (0.5 nmol/h) via osmotic minipumps for 23 days. Mice were fasted overnight and treated with insulin (3 U/kg body wt) for 5 min, and liver extracts were immunoprecipitated with antibodies against IR and IRS1 and immunoblotted with

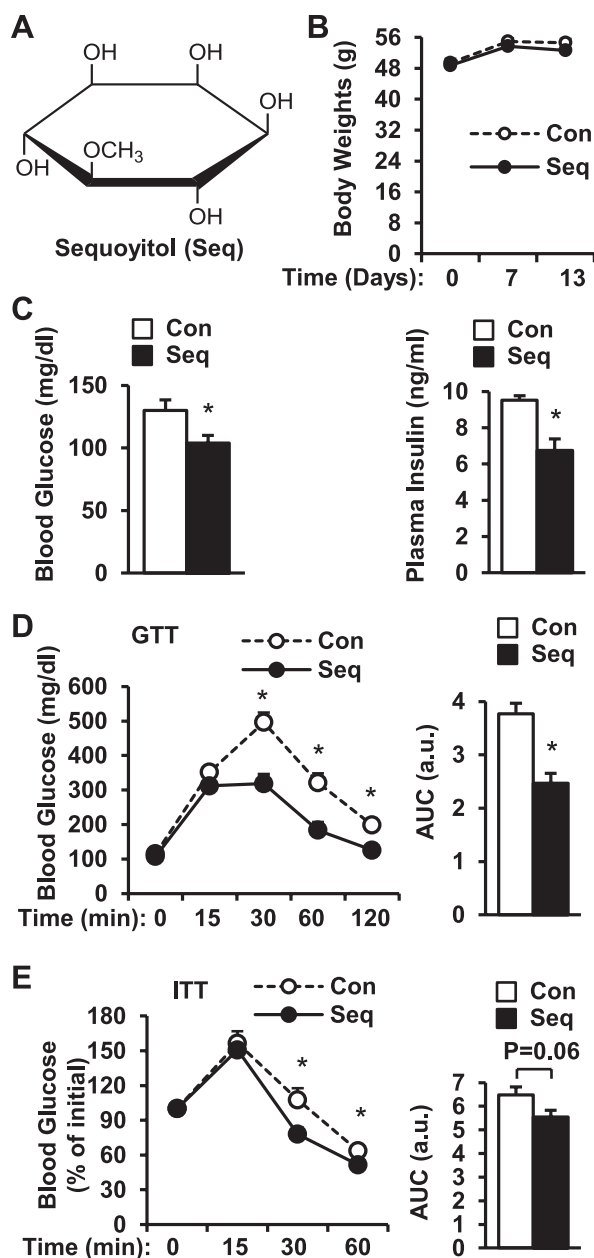


Fig. 1. Subcutaneous administration of sequoyitol decreases blood glucose and plasma insulin and improves glucose intolerance in *ob/ob* mice. **A**: sequoyitol structure. **B**: *ob/ob* males (10 wk) were subcutaneously administered with sequoyitol (0.5 nmol/h) ($n = 7$) or a saline vehicle ($n = 6$) via osmotic minipumps. Body weights were monitored daily. **C**: fasting blood glucose and plasma insulin levels 14 days after sequoyitol treatments. **D**: glucose tolerance test (GTT). Mice (17 days after treatments) were fasted for 16 h and intraperitoneally injected with D-glucose (0.8 g/kg body wt). Blood glucose levels were monitored. a.u., Arbitrary units; AUC, area under the curve. **E**: insulin tolerance test (ITT). Mice (10 days after treatments) were fasted for 5 h and intraperitoneally injected with insulin (4 U/kg body wt). Blood glucose levels were measured and normalized to initial values. Con, control. Values are means $\pm$ SE. * $P < 0.05$.

anti-phosphotyrosine (α -PY), IR, or IRS1 antibodies. In parallel experiments, liver extracts were immunoblotted with the indicated antibodies. In vehicle-treated control mice, insulin stimulated tyrosine phosphorylation of IR, modestly stimulated tyrosine phosphorylation of IRS1 and Ser⁴⁷³/Thr³⁰⁸ phospho-

rylation of Akt, but did not stimulate ERK phosphorylation, in agreement with insulin resistance in these mice (Fig. 2A). Sequoyitol treatments significantly increased the ability of insulin to stimulate Akt phosphorylation (Fig. 2A). Ser⁴⁷³ and Thr³⁰⁸ phosphorylation was 68 and 118% higher in sequoyitol-

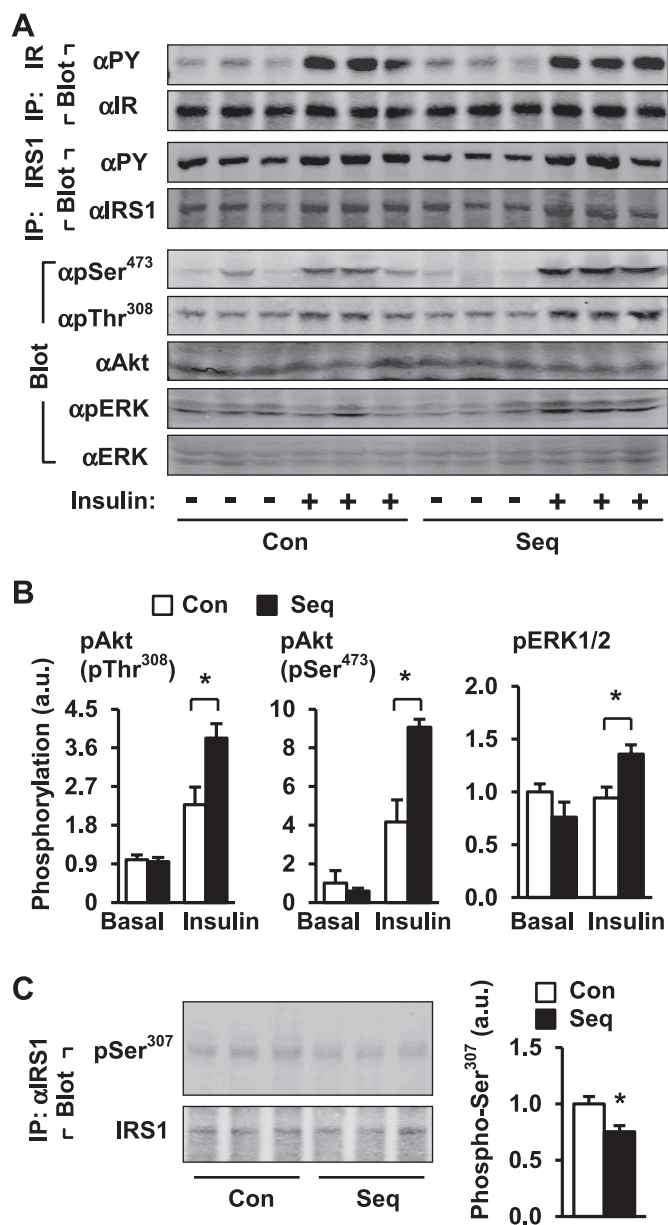


Fig. 2. Sequoyitol improves insulin signaling in *ob/ob* mice. The *ob/ob* mice (10 wk) were subcutaneously administered with sequoyitol (0.5 nmol/h) or saline via osmotic minipumps. Twenty-three days after treatments, mice were fasted overnight and treated with insulin (3 U/kg body wt for 5 min) via inferior vena. **A**: liver extracts were immunoprecipitated with anti-insulin receptor (IR) and anti-IR substrate-1 (IRS1) antibodies and immunoblotted with anti-phosphotyrosine (α -PY), IR, or IRS1 antibodies. In parallel, liver extracts were immunoblotted with anti-phospho-Akt (pAkt) (α -pSer⁴⁷³ or α -pThr³⁰⁸), Akt, phospho-ERK (pERK), and ERK antibodies. IP, immunoprecipitated. **B**: Akt or ERK phosphorylation was quantified using densitometry and normalized to total Akt or ERK levels (Con: $n = 3$; sequoyitol: $n = 3$). **C**: liver extracts (from the insulin-treated groups) were immunoprecipitated with anti-IRS1 antibody and immunoblotted with anti-phospho-IRS1 (pIRS1) (pSer³⁰⁷) or anti-IRS1 antibodies. Phosphorylation of Ser³⁰⁷ was quantified and normalized to total IRS1 levels. Values are means $\pm$ SE. * $P < 0.05$.

treated mice than in vehicle-treated mice, respectively (Fig. 2B). Sequoyitol also modestly increased insulin-stimulated tyrosine phosphorylation of IRS1 and ERK phosphorylation (Figs. 2, A and B). Ser³⁰⁷ phosphorylation of IRS1 is believed to negatively regulate insulin sensitivity (2, 33). To measure Ser³⁰⁷ phosphorylation, mice were treated with saline or sequoyitol for 23 days as described above, and liver extracts were immunoprecipitated with anti-IRS1 antibody and immunoblotted with anti-phospho-IRS1 (pSer³⁰⁷) antibody. Sequoyitol treatments significantly decreased Ser³⁰⁷ phosphorylation of IRS1 (Fig. 2C). These data suggest that sequoyitol acts as an insulin sensitizer to improve glucose metabolism in *ob/ob* mice.

Oral administration of sequoyitol improves insulin sensitivity and glucose metabolism in *ob/ob* mice. To verify the anti-diabetic effect of sequoyitol, *ob/ob* male and female mice (8–9 wk) were administered with sequoyitol (40 mg/kg body wt, twice daily) by oral gavage. Sequoyitol treatments (17 days) did not alter body weight in either males (control: 45.7 ± 0.7 g, *n* = 5; sequoyitol: 47.4 ± 0.6 g, *n* = 5; *P* = 0.1026) or females (control: 44.7 ± 1.1 g, *n* = 5; sequoyitol: 43.9 ± 1.3 g, *n* = 6; *P* = 0.6286). As expected, sequoyitol treatments (18 days) reduced blood glucose in both males and females (Fig. 3A). Oral administration of sequoyitol also markedly improved glucose intolerance during GTT (Fig. 3B). Surprisingly, plasma insulin levels were similar between control and treated groups in both males (control: 4.4 ± 0.4 ng/ml, *n* = 5; sequoyitol: 5.1 ± 0.1 ng/ml, *n* = 5; *P* = 0.173) and females (control: 4.9 ± 0.1 ng/ml, *n* = 5; sequoyitol: 4.7 ± 0.2 ng/ml, *n* = 6; *P* = 0.483). These data suggest that blood insulin levels do not contribute to the hypoglycemic effect of sequoyitol under these conditions. In ITT, blood glucose levels were significantly lower 15, 30, and 60 min after insulin injection in both males and females (Fig. 3C). These results suggest that sequoyitol has a potential to be used as an oral diabetes medication to improve hyperglycemia and glucose intolerance.

Sequoyitol directly enhances insulin signaling and suppresses glucose production in hepatocytes. To determine whether sequoyitol directly targets hepatocytes, we examined the effect of sequoyitol on insulin signaling in HepG2 cells, human hepatoblastoma cells. HepG2 cells were pretreated with or without sequoyitol and subsequently stimulated with insulin. Cell extracts were immunoprecipitated with anti-IRS1 antibody and immunoblotted with α-PY antibody. Cell extracts were also immunoblotted with anti-phospho-Akt (pSer⁴⁷³) antibody. Insulin stimulated phosphorylation of IRS1 and Akt in control cells, and sequoyitol further enhanced insulin-stimulated phosphorylation of both IRS1 and Akt (Fig. 4A). We quantified IRS1 and Akt phosphorylation and observed that sequoyitol increased insulin-stimulated phosphorylation of IRS1 by 114% and Akt by 54% (Fig. 4B). Because tumor necrosis factor (TNF)α contributes to insulin resistance in obesity (1, 15, 16, 30, 37, 38), we examined the possibility that sequoyitol may counteract TNFα-induced insulin resistance. HepG2 cells were pretreated with TNFα in the presence or absence of sequoyitol, and the pretreated cells were stimulated with insulin. TNFα suppressed insulin-stimulated tyrosine phosphorylation of IRs and IRS1 and Akt phosphorylation as expected (Fig. 4C, lanes 5 vs. 2). Sequoyitol significantly increased insulin-stimulated phosphorylation of IR (by 46%), IRS1 (by 48%), and Akt (pSer⁴⁷³) (by 61%) (Fig. 4C). We also

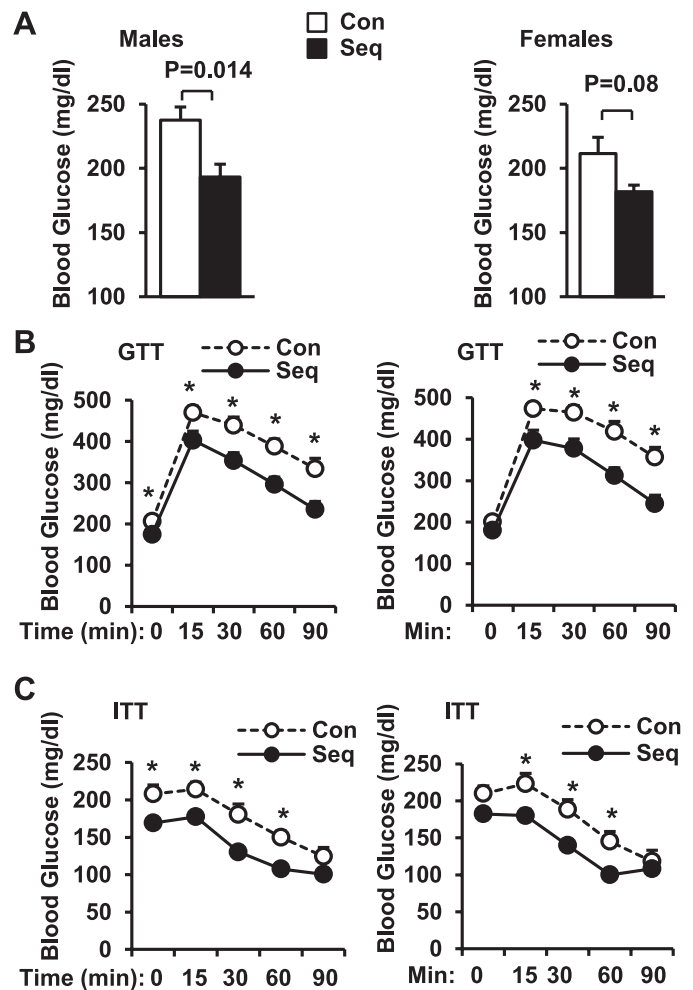
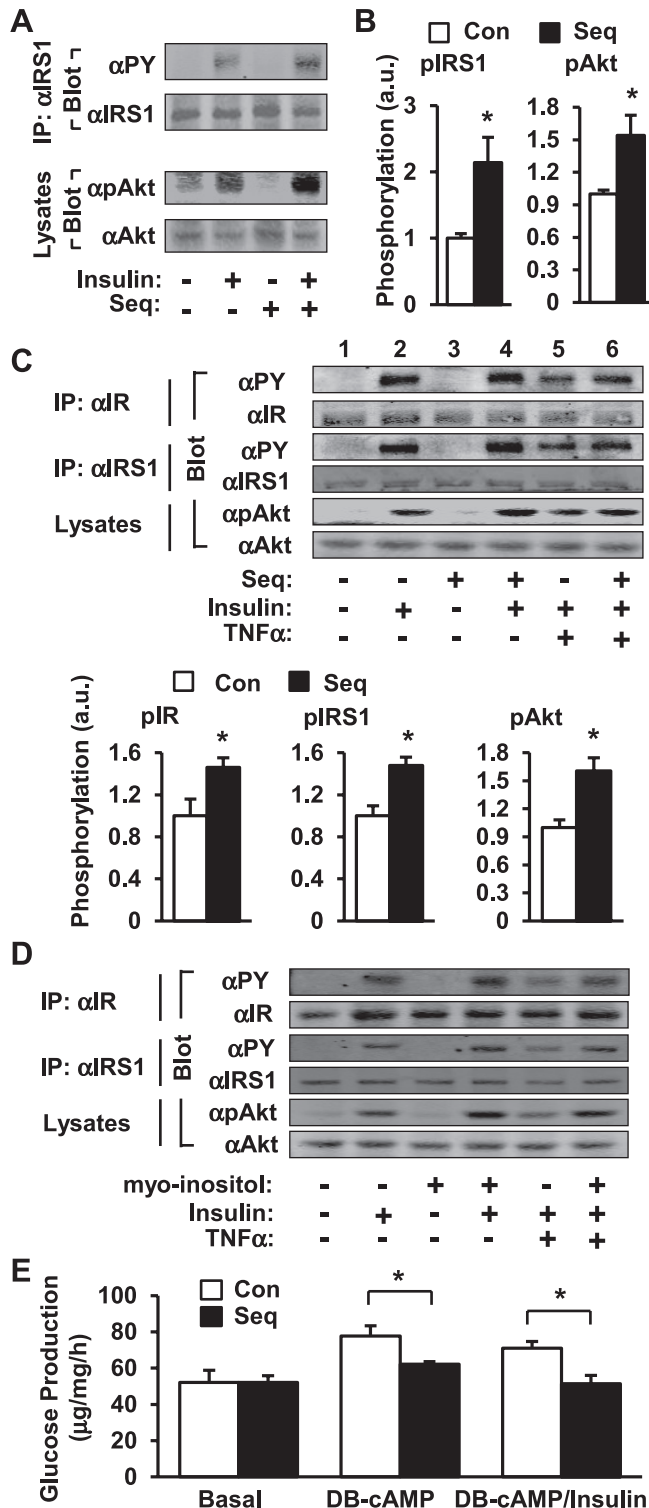


Fig. 3. Oral administration of sequoyitol improves hyperglycemia and glucose intolerance in *ob/ob* mice. The *ob/ob* mice (8–9 wk) were administered by oral gavage without (males: *n* = 5; females: *n* = 5) or with sequoyitol (40 mg/kg body wt, twice daily; males: *n* = 5; females: *n* = 6). A: fasting (6 h) blood glucose 18 days after treatments. B: GTT. Mice (31 days after treatments) were fasted for 6 h and intraperitoneally injected with D-glucose (0.75 g/kg body wt). Blood glucose levels were monitored. C: ITT. Mice (27 days after treatments) were fasted for 6 h and intraperitoneally injected with insulin (1.5 U/kg body wt). Blood glucose levels were measured. Values are means ± SE. **P* < 0.05.

examined the effect of *myo*-inositol, a potential derivative of sequoyitol in vivo, on insulin signaling in hepatocytes. Like sequoyitol, *myo*-inositol pretreatments also improved insulin-stimulated tyrosine phosphorylation of IR and IRS1 and Ser⁴⁷³ phosphorylation of Akt and counteracted TNFα inhibition of insulin signaling (Fig. 4D). These data suggest that sequoyitol and its derivatives sensitize insulin responses in hepatocytes.

To determine whether sequoyitol suppresses hepatic glucose production, primary hepatocytes were pretreated without or with sequoyitol (100 μM) for 16 h and subsequently subjected to glucose production assays. Basal glucose production was similar between control and sequoyitol-pretreated cells (Fig. 4E). DB-cAMP, a cAMP analog, stimulated glucose production by 49% in control cells, but only by 19% in sequoyitol-pretreated hepatocytes (Fig. 4E). These data indicate that sequoyitol is able to directly suppress hepatic glucose production in an insulin-independent manner. Either sequoyitol or

insulin alone partially suppressed glucose production in DB-cAMP-stimulated hepatocytes; however, sequoyitol and insulin in combination completely inhibited DB-cAMP-stimulated glucose production (Fig. 4E). These data suggest that sequoyitol and insulin act additively or synergistically to suppress hepatic glucose production.



Sequoyitol directly enhances insulin signaling and glucose uptake in adipocytes. To determine whether sequoyitol directly targets adipocytes, 3T3-L1 adipocytes were pretreated with or without sequoyitol before insulin stimulation. Insulin stimulated tyrosine phosphorylation of IR and IRS1 in control cells, and sequoyitol further increased insulin-stimulated phosphorylation of IR by 31% and IRS1 by 73% (Fig. 5, A and B). To determine whether sequoyitol counteracts TNF α inhibition of insulin signaling, 3T3-L1 adipocytes were pretreated with TNF α in the presence or absence of sequoyitol. TNF α inhibited insulin signaling in control cells as expected (Fig. 5C, lanes 5 vs. 2). Sequoyitol increased insulin-stimulated phosphorylation of IR by 53%, IRS1 by 69%, and Akt by 74% in TNF α -treated 3T3-L1 adipocytes (Fig. 5C). Similarly, myo-inositol, a potential sequoyitol derivative, also enhanced insulin-stimulated phosphorylation of IR, IRS1, and Akt and counteracted TNF α inhibition of insulin signaling in adipocytes (Fig. 5D). These data suggest that sequoyitol and its derivatives directly enhance insulin sensitivity in adipocytes.

To determine whether sequoyitol regulates glucose uptake, 3T3-L1 adipocytes were pretreated with sequoyitol for 6 or 12 h before insulin stimulation. Insulin stimulated glucose uptake in control adipocytes as expected, and sequoyitol further increased insulin-stimulated glucose uptake by 34 and 81% 6 and 12 h after pretreatments, respectively (Fig. 5E). These data provide additional evidence supporting sequoyitol as an insulin sensitizer. Prolonged sequoyitol treatments (12 h) alone also increased basal glucose uptake (Fig. 5E). In agreement, sequoyitol increased both basal and insulin-stimulated glucose uptake in mouse primary adipocytes (Fig. 5F). These data suggest that sequoyitol promotes glucose uptake into adipocytes by both insulin-dependent and insulin-independent mechanisms.

Sequoyitol protects islet β -cells against oxidative injury. To determine whether sequoyitol directly targets β -cells, INS-1 cells, derived from rat β -cells, were pretreated without (control) or with sequoyitol before STZ (0.5 mM) or H₂O₂ (10 μ M)

Fig. 4. Sequoyitol enhances insulin signaling and suppresses glucose production in hepatocytes. **A:** HepG2 cells were pretreated without or with sequoyitol (100 μ M) for 12 h before insulin (10 nM) stimulation (5 min). Cell extracts were immunoprecipitated with α -IRS1 and immunoblotted with α -PY. The blots were reprobed with α -IRS1. Cell extracts were also immunoblotted with α -pSer473 or α -Akt. **B:** pIRS1 and Akt were quantified by densitometry and normalized to total IRS1 and Akt levels, respectively (Con: $n = 3$; sequoyitol: $n = 3$). **C:** HepG2 cells were treated with or without sequoyitol (100 μ M) in the presence or absence of TNF α (10 ng/ml) for 12 h and then stimulated with insulin (10 nM for 5 min). Cell extracts were immunoprecipitated with α -IR and α -IRS1 and immunoblotted with α -PY. The same blots were reprobed with α -IR or α -IRS1, respectively. Cell extracts were also immunoblotted with α -pSer473 or α -Akt. IR, IRS1, and Akt phosphorylation was quantified and normalized to total IR, IRS1, and Akt protein levels, respectively (Con: $n = 3$; sequoyitol: $n = 3$). **D:** HepG2 cells were pretreated with myo-inositol (100 μ M) in the presence or absence of TNF α (10 ng/ml) for 12 h and then stimulated with insulin (10 nM for 5 min). Cell extracts were immunoprecipitated with α -IR and α -IRS1, and immunoblotted with α -PY, α -IR, or α -IRS1, respectively. Cell extracts were also immunoblotted with α -pSer473 or α -Akt. **E:** primary hepatocytes were prepared from C57BL/6 males, treated without or with sequoyitol (100 μ M) overnight, stimulated with a vehicle, $N^6,2'$ -O-dibutyryl adenosine 3',5'-cyclic monophosphate sodium salt (DB-cAMP; 10 μ M), or DB-cAMP plus insulin (100 nM), and subjected to glucose production assays. Glucose production was normalized to total hepatocyte protein levels (Con: $n = 4$; sequoyitol: $n = 4$). Each experiment was performed three times. Values are means $\pm$ SE. * $P < 0.05$.

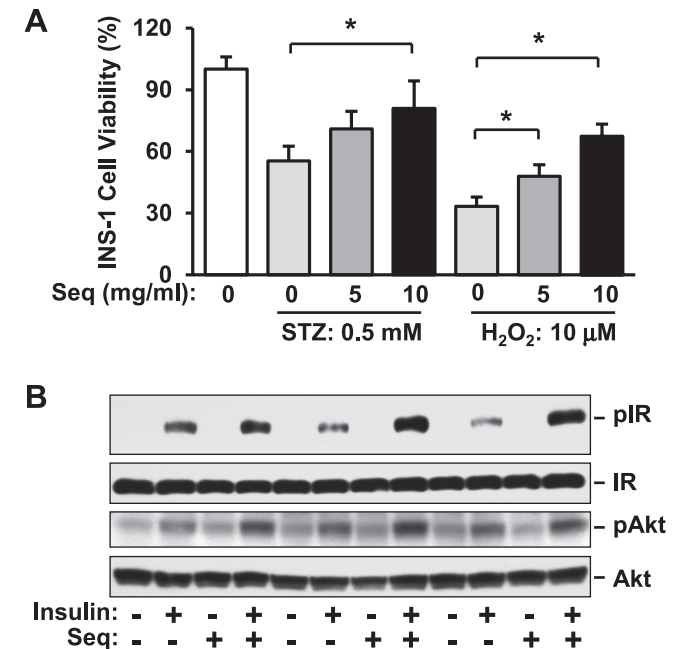
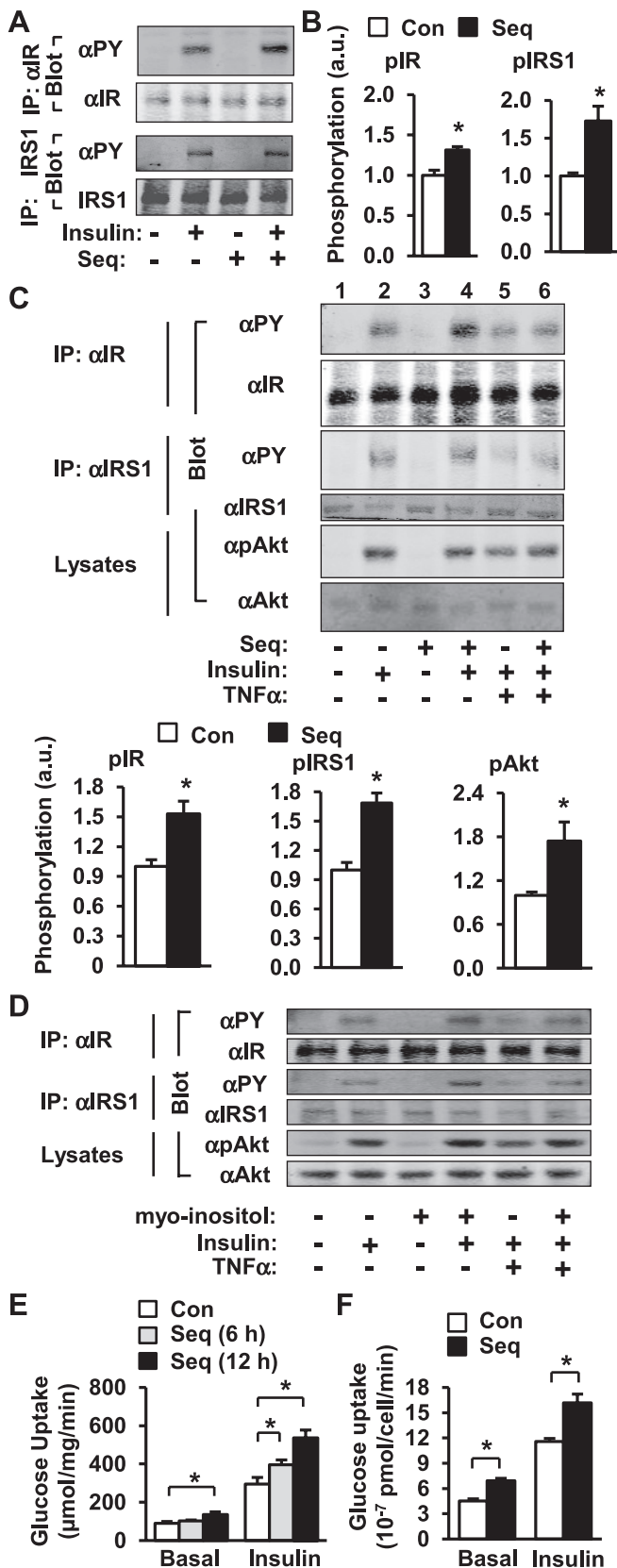


Fig. 6. Sequoyitol protects against streptozotocin (STZ)- and H₂O₂-induced INS-1 β-cell death. **A**: INS-1 β-cells were incubated with sequoyitol for 3 h and then with STZ or H₂O₂ in the presence of sequoyitol for an additional 12 h. Cell viability was measured using MTT assays ($n = 4$ for each group). **B**: INS-1 cells were deprived of serum overnight, pretreated with or without 10 mg/ml sequoyitol for 6 h, and then stimulated with 50 nM insulin for 15 min. Cell extracts were immunoblotted with α-PY, α-IR, α-pSer473, or α-Akt. Values are means $\pm$ SE. * $P < 0.05$.

treatments, and cell viability was measured using MTT assays. Both STZ and H₂O₂ markedly reduced the viability of control INS-1 cells, and sequoyitol dose-dependently increased the viability of STZ- and H₂O₂-treated INS-1 cells (Fig. 6A). These data indicate that sequoyitol protects against β-cell injury and death under oxidative stress conditions. To determine whether sequoyitol improves insulin signaling in INS-1

Fig. 5. Sequoyitol enhances insulin signaling and increases glucose uptake in adipocytes. **A**: 3T3-L1 preadipocytes were fully differentiated into adipocytes (day 8). The adipocytes were pretreated without or with sequoyitol (100 μM) for 12 h and then stimulated with insulin (10 nM) for 5 min. Cell extracts were immunoprecipitated with α-IR or α-IRS1 and immunoblotted with α-PY. The same blots were reprobed with α-IR or α-IRS1. **B**: pIRS1 and Akt were quantified by densitometry and normalized to total IRS1 and Akt levels, respectively (Con: $n = 3$; sequoyitol: $n = 3$). **C**: 3T3-L1 adipocytes were treated with or without sequoyitol (100 μM) in the presence or absence of TNFα (10 ng/ml) for 12 h and then stimulated with insulin (10 nM) for 5 min. Cell extracts were immunoprecipitated with α-IR and α-IRS1 and immunoblotted with α-PY, α-IR, or α-IRS1, respectively. Cell extracts were also immunoblotted with α-pSer473 or α-Akt. IR, IRS1, and Akt phosphorylation was quantified and normalized to total IR, IRS1, and Akt protein levels, respectively (Con: $n = 3$; sequoyitol: $n = 3$). **D**: 3T3-L1 adipocytes were treated with or without myo-inositol (100 μM) in the presence or absence of TNFα (10 ng/ml) for 12 h and then stimulated with insulin (10 nM) for 5 min. Insulin signaling was examined as described in **C**. **E**: 3T3-L1 adipocytes were pretreated without or with sequoyitol (100 μM) for 6 or 12 h. The cells were subsequently stimulated with or without insulin (10 nM) and subjected to glucose uptake assays. **F**: primary adipocytes were isolated from C57BL/6 males fed a high-fat diet for 8 wk, treated without or with sequoyitol (100 μM) for 3 h, stimulated with or without insulin (5 nM), and subjected to glucose uptake assays. Each experiment was performed three times. Values are means $\pm$ SE. * $P < 0.05$.

cells, INS-1 cells were pretreated with or without 10 mg/ml sequoyitol for 6 h before insulin stimulation (50 nM for 15 min). Sequoyitol pretreatments increased the ability of insulin to stimulate phosphorylation of both IR and Akt (pSer⁴⁷³) in three repeated experiments (Fig. 6B). To determine whether sequoyitol improves β -cell function in animals, C57BL/6J males were injected with STZ, a β -cell toxin, and divided into control and sequoyitol-treated groups. Sequoyitol was administered in drinking water (70–100 mg·kg⁻¹·day⁻¹). Body weights were similar between these two groups during the treatments (Fig. 7A). STZ treatments progressively increased blood glucose in the control group, and sequoyitol markedly attenuated STZ-induced hyperglycemia in the sequoyitol-treated group (Fig. 7B). Blood glucose levels decreased by 26% 31 days after sequoyitol treatments. Sequoyitol also increased plasma insulin levels by 155% (Fig. 7C) and significantly improved glucose intolerance (Fig. 7D). These data raise the possibility that sequoyitol may protect against β -cell injury.

DISCUSSION

Herbal extracts have been reported to reduce blood glucose in animals (6, 8, 17, 22, 41); however, these extracts are extremely complex, and the leading compounds that exert antidiabetic effects remain poorly understood. The cellular

targets of these herbal extracts and the molecular basis of their actions are unknown. Lack of these important information limits our ability to use herbal therapies to treat diabetes. In this work, we have identified sequoyitol, a natural compound present in many plants (e.g., *Amentotaxus yunnanensis*, *Aristolochia arcuata*, and *Crossostephium Chinese*) (12, 24, 43), as a novel antidiabetic small molecule (molecular weight: 194).

We demonstrated that chronic administration of sequoyitol significantly improved hyperglycemia and glucose intolerance in *ob/ob* mice. Sequoyitol treatments did not alter body weights, indicating that the improvement in metabolism is unlikely to be secondary to an alteration in energy balance and adiposity. In agreement, sequoyitol not only suppressed glucose production in primary hepatocytes, but also promoted glucose uptake in both 3T3-L1 adipocytes and primary adipocytes. These observations suggest sequoyitol attenuates hyperglycemia and glucose intolerance in type 2 diabetes, at least in part by directly targeting hepatocytes and adipocytes and improving the metabolic function of these two cell types. Sequoyitol is water soluble; additionally, both oral and subcutaneous administration of sequoyitol improves hyperglycemia and glucose intolerance. These properties make sequoyitol an appealing oral diabetes medication candidate.

To gain insight into the molecular mechanisms of sequoyitol action, we examined insulin signaling. We observed that sequoyitol directly enhanced insulin-stimulated phosphorylation of IR, IRS1, and Akt in hepatocytes, adipocytes, and/or INS-1 β -cells. It also counteracted TNF α inhibition of insulin signaling in hepatocytes and adipocytes. Subcutaneous administration of sequoyitol improved insulin signaling in *ob/ob* mice. Chronic sequoyitol treatments reduced hyperinsulinemia and increased the hypoglycemic effect of insulin in these mice. These observations suggest that sequoyitol ameliorates hyperglycemia in part by functioning as an insulin sensitizer. The direct molecular targets of sequoyitol are currently unknown. Sequoyitol may serve as a precursor of inositols, phosphatidylinositols, and other inositol derivatives, and these inositol derivatives may mediate sequoyitol's antidiabetic effect. Indeed, like sequoyitol, *myo*-inositol enhanced insulin signaling in both hepatocytes and adipocytes.

Sequoyitol alone suppressed glucose production in primary hepatocytes and stimulated basal glucose uptake in adipocytes, suggesting that sequoyitol is able to improve hyperglycemia and glucose intolerance in *ob/ob* mice by an insulin-independent mechanism. Interestingly, sequoyitol alone does not stimulate Akt phosphorylation, suggesting that the Akt pathway may not mediate the sequoyitol action under these conditions. IPGs, endogenous inositol derivatives, are believed to act as intracellular mediator of insulin metabolic action (7, 21, 23, 29, 36). IPGs appear to regulate glucose metabolism by activating pyruvate dehydrogenase phosphatases and protein phosphatase 2C (5, 13, 23, 25, 27). Sequoyitol (5-*O*-methyl-*myo*-inositol) may promote IPG production by serving as a precursor or a regulator. Additionally, hyperglycemia is associated with both deficiency of *chiro*-inositol and increased *myo*-inositol-to-*chiro*-inositol ratios (3, 7, 20, 23). Sequoyitol may promote *chiro*-inositol production and decrease *myo*-inositol-to-*chiro*-inositol ratios, thus improving hyperglycemia and glucose intolerance in mice with obesity.

Sequoyitol also improved hyperglycemia and glucose intolerance in STZ-treated mice. STZ promotes hyperglycemia and

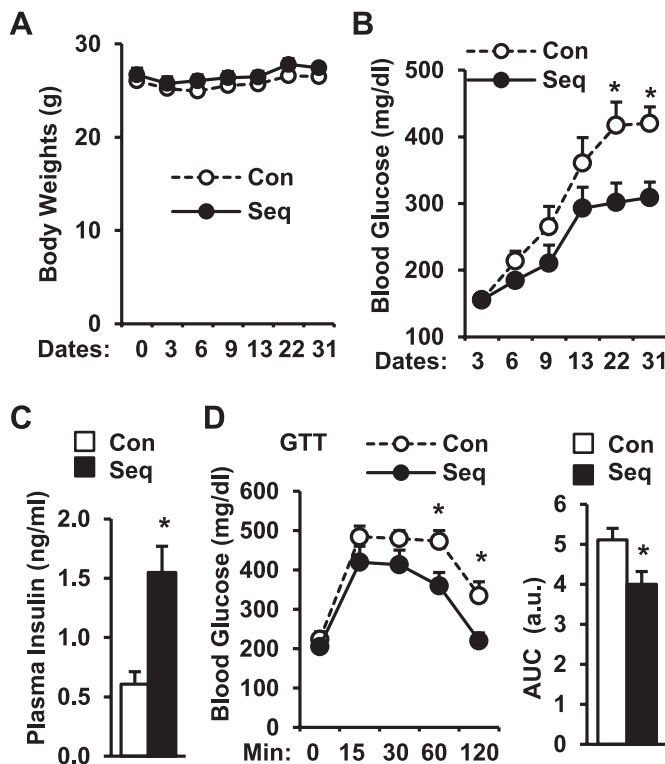


Fig. 7. Sequoyitol decreases hyperglycemia and glucose intolerance and increases plasma insulin in STZ-treated mice. C57BL/6 males (9 wk) were intraperitoneally injected with STZ (160 mg/kg body wt) and randomly divided into control and sequoyitol-treated groups. Con ($n = 8$) mice had free access to water; sequoyitol ($n = 8$) mice had free access to water supplemented with sequoyitol (7 mg/ml). A: growth curves. B: randomly fed blood glucose (10:00–11:00 AM). C: plasma insulin levels 14 days after sequoyitol treatments (Con: $n = 7$; sequoyitol: $n = 7$). D: GTT (glucose: 0.6 g/kg body wt) 29 days after sequoyitol treatments. Values are means $\pm$ SE. * $P < 0.05$.

glucose intolerance by destroying islet β -cells. Sequoyitol treatments markedly increased plasma insulin levels (by 155%), suggesting that sequoyitol preserves β -cell viability and/or function in vivo. In agreement, sequoyitol enhanced insulin signaling in INS-1 cells and directly protected against STZ- and H_2O_2 -induced INS-1 β -cell death. In supporting these findings, a herbal extract containing sequoyitol has been reported to improve insulin secretion from rat islets (45). The molecular basis of sequoyitol's cytoprotection is unclear. D-Pinitol, another methyl derivative of inositol, acts as an antioxidant to protect against hepatocyte injury (35). Sequoyitol may similarly protect β -cells from oxidative injury by acting as an antioxidant.

In summary, we have identified sequoyitol as a new herbal constituent with an antidiabetic property. Both oral and subcutaneous administrations of sequoyitol improve hyperglycemia and glucose intolerance in both *ob/ob* and STZ-treated mice. Sequoyitol directly improves glucose metabolism in hepatocytes and adipocytes by both insulin-dependent and insulin-independent mechanisms. It also protects β -cells from oxidative injury. Thus sequoyitol has a therapeutic potential and may serve as an oral diabetes medication.

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DISCLOSURES

T. Wang is an employee of Xiangbei Welman Pharmaceutical Co., Ltd., which provides sequoyitol.

AUTHOR CONTRIBUTIONS

H.S., M.S., K.W.C., S.W., Z.C., and L.S. performed experiments; H.S., M.S., K.W.C., S.W., Z.C., L.S., Y.L., and L.R. analyzed data; H.S., M.S., Y.L., and L.R. interpreted results of experiments; H.S., M.S., and L.R. prepared figures; H.S., Y.L., and L.R. drafted manuscript; H.S., M.S., K.W.C., S.W., Z.C., L.S., T.W., Y.L., and L.R. approved final version of manuscript; M.S., Y.L., and L.R. edited and revised manuscript; Y.L. and L.R. conception and design of research.

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